

Tomás David Navarro Múnera

DEVOPS ENGINEER · SECURITY RESEARCHER

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Summary

DevOps Engineer and Security Researcher who graduated with an M.A.Sc. in Information System Security from Concordia University, with a research focus on container security. Experienced in automating and securing AWS infrastructure with Terraform, Ansible, Docker, and Kubernetes, and in researching kernel and cross-container attacks as a Mitacs-funded Graduate Research Student with the NSERC/Ericsson IRC in SDN/NFV Security. Computer Science graduate from EAFIT University with a growing focus on offensive security, gained through CTF competitions and hands-on penetration testing.

Education

Concordia University

M.A.Sc. IN INFORMATION SYSTEM SECURITY

Montreal, Quebec, Canada

May 2023 - Aug. 2026

EAFIT University

B.S. IN COMPUTER SCIENCE AND ENGINEERING, GPA: 4.28/5

Medellín, Colombia

Jan. 2018 - Dec. 2022

Experience

Concordia University

GRADUATE RESEARCH STUDENT - NSERC/ERICSSON IRC IN SDN/NFV SECURITY

Montreal, Quebec, Canada · Hybrid

May. 2023 - Aug. 2026

- Selected for a Mitacs Accelerate research internship (Sep. - Dec. 2023) with Ericsson.
- Use of Docker and Podman for testing cross-container vulnerabilities and exploits.
- Use of eBPF for detecting kernel attacks and mitigation solutions.
- Research Kernel and cross-container attacks.

Pyxis SAS

DEVOPS ENGINEER

Medellín, Colombia

May. 2022 - March. 2023

- Use of Terraform and Terragrunt for IaC construction and deployment in AWS.
- Use of Ansible as an automation tool for deployment of server tools and configuration.
- AWS Networking and Configuration for High Availability and High Scalability in applications.
- Use of Docker for environmental configuration and deployment of applications in AWS.

Empresas Públicas de Medellín

DATA SCIENTIST INTERN

Medellín, Colombia

Aug. 2021 - Feb 2022

- Use of Data Science programming languages [Python, Matlab, R].
- Build Machine Learning Models for hydrological forecasting.
- Data Analysis of the models' results.
- Build WebScraping Tool to extract data from XM.com.co.
- Build API connector tool to get data from XM.com.co to analyse data.

EAFIT University, Apolo Supercomputing Center

RESEARCH ASSISTANT

Medellín, Colombia

Sep. 2019 - Jun. 2021

- Linux System Admin, review packages for vulnerabilities.
- Deployment of private web-apps in Docker to secure the cluster.
- Created "LMOD module generator" script for apps usage [Bash].
- Use of Ansible and Vagrant as an automation tool for deployment of server tools and configuration.
- Install HPC applications for cluster users.

Publications

KubeBPF: eBPF-Based In-Kernel Machine Learning Approach for Detecting and Mitigating Abstract Resource Attacks for Containers

M.A.Sc. THESIS · CONCORDIA UNIVERSITY

Aug. 2026

Accepted

CONnecting The EXtra doTS (CONTEXTS): Correlating External Information about Points of Interest for Attack Investigation

SAREH MOHAMMADI, HUGO KERMABON-BOBINNEC, AZADEH TABIBAN, LINGYU WANG, TOMÁS NAVARRO MÚNERA, YOSR

JARRAYA

IEEE S&P 2025

Accepted

Skills

DevOps AWS, Docker, Vagrant, Linux, eBPF, Ansible, Kubernetes, Terraform, Terragrunt

Back-end Laravel, Django, REST API

Front-end Bootstrap, HTML5

Programming Python, JAVA, LaTeX, C++, C, PHP, R, Matlab, GTK3

Languages Spanish (Native), English (Fluent), French (Basic)

Extracurricular Activity

Various CTFs

PARTICIPANT

- Participated in NorthSec 2024 and NorthSec 2025 CTFs.
- Participated in PolyPwn 2025 and 2026 CTF (Third place in the 2026 edition).
- Participated in @Hack 2024 CTF.

Montreal, Canada

Feb. 2024 - Present

AtHack CTF

ATHACK ORGANIZER

- Helped organize @Hack CTF event where more than 300 participated.
- Gave support on challenges to participants.

Montreal, Canada

Feb. 2025

Courses & Certificates

Cloud and DevOps Fundamentals. Cloud and Automation tools.

CERTIFICATE

EPAM Learning

Apr. 2025

AWS Certified Cloud Practitioner

CERTIFICATE

Amazon Web Services

May. 2023

Complete Ethical Hacking Bootcamp 2022: Zero to Mastery

CERTIFICATE

Udemy

Jun. 2022

DevOps in the Cloud with Terraform, Ansible, and Jenkins

CERTIFICATE

Udemy

Jun. 2022

- Deployed Grafana and Prometheus to EC2 instances using DevOps tools Terraform, Ansible, and Jenkins.